

Automated Smart Meal Planner:
An AI-Based Nutrition Tracker and Meal Planner

Maxzine Rossalyn Reid
Senior Capstone Project – Information Technology
University of the District of Columbia
Professor Byunggu Yu, Ph.D.
October 2025

Maxzine Reid
University of the District of Columbia
Information Technology – Senior Capstone Project
October 2025

Professor Byunggu Yu, Ph.D.
University of the District of Columbia

Dear Professor Yu,

My name is Maxzine Reid, and I am a senior Information Technology student at the University of the District of Columbia. Throughout my studies, I've worked to blend creativity with practical technical solutions. Projects like my Change Management Application and my growing experience in Python have helped me understand how to turn ideas into meaningful, user-centered tools.

For my capstone, I am proud to present the Automated Smart Meal Planner, an AI-based nutrition assistant that uses natural language processing and external APIs to create personalized meal plans, generate grocery lists, and encourage healthy eating habits. The chatbot is designed using Google Dialogflow and the Spoonacular API, and I am developing a fallback system using TheMealDB and Open Food Facts to ensure long-term reliability.

After recovering from illness earlier this semester, I have regained steady progress on this project. I am currently testing live API responses and building conversational logic that adapts to user preferences. My goal is to deliver a fully functional, accessible health assistant prototype by finals week. Thank you for your continued guidance and understanding as I worked to get back on track.

Sincerely,
Maxzine Rossalyn Reid

Audience Analysis

Gatekeeper:

Professor Byunggu Yu, Ph.D. – The project supervisor and course instructor. Professor Yu will evaluate the technical quality, academic rigor, and completeness of the proposal. His approval is required for the project to advance to the final phase.

Primary Audience:

Capstone Evaluation Committee – Faculty reviewers in the Information Technology Department. They will assess how effectively the proposal demonstrates technical competence, problem-solving, and research-based planning.

Secondary Audience:

End Users – Health-Conscious Individuals. These users include students, working adults, or individuals managing diets for health reasons. They will be most interested in how simple, accurate, and practical the chatbot feels.

Auxiliary Audience:

Future Employers and Internship Reviewers. While not directly involved in grading, they may review the project as evidence of my ability to plan, document, and implement real-world technology solutions.

Table of Contents

Cover Letter	2
Audience Analysis	3
Title Page	1
Table of Contents	4
I. Introduction and Context	5
II. Literature Review and Available Models	7
III. Analysis and Synthesis	9
IV. Proposed Solution	11
Implementation Progress (as of October 2025)	13
V. Current Status and Work Plan	15
VI. References	17
VII. Appendices	18
Conclusion and Reflection	22

I. Introduction and Context

Eating healthy in a fast-paced world is difficult for many people. Between busy schedules, conflicting dietary advice, and complicated food apps, maintaining good nutrition often feels like an overwhelming task. Many available meal-tracking tools provide calorie counts but lack guidance, personalization, or encouragement. The challenge isn't just knowing what to eat, it's having a reliable, intelligent system that helps users make better choices day by day.

The Automated Smart Meal Planner addresses this challenge through artificial intelligence. This project combines chatbot technology, nutrition databases, and recipe APIs to help users plan meals, generate grocery lists, and stay on track with their health goals. The chatbot acts as a friendly, supportive assistant rather than a static app, offering simple, conversational guidance that grows smarter over time.

My inspiration for this project came from personal experience. I've used various fitness and diet apps that promised to simplify healthy living but felt repetitive, confusing, or impersonal. They often required users to log every detail manually. This left me asking: what if a chatbot could handle the heavy lifting like finding recipes, generating meal plans, and adapting automatically to preferences or dietary needs?

From a professional standpoint, this project reflects my academic journey through software engineering, system design, and cybersecurity. I learned structured system development in Software Engineering (CSCI 341) and practical coding in Python and Web Technologies. I'm

applying these skills to build an intelligent system that merges API integration, data handling, and user interaction.

Healthy eating is not only a personal issue, it's a global one. According to the World Health Organization (2022), unhealthy diets are linked to major health risks, including obesity, diabetes, and heart disease. Yet, many people lack accessible, personalized tools that fit their lifestyles. With AI and natural language processing, I can create a tool that meets users where they are being guided toward balance, convenience, and healthier living.

This project's long-term vision is to create a platform that bridges technology and wellness where data-driven insights and user empathy come together.

II. Literature Review and Available Models

Several AI-powered tools attempt to solve nutrition and meal-planning challenges, but most still fall short of providing true personalization and adaptability.

MyFitnessPal is one of the most popular health apps, allowing users to log meals and track calories. However, it lacks interactive features and intelligent guidance. As noted by Healthline (2023), it provides useful information but can feel repetitive and uninspired for long-term engagement.

Lifesum offers a more visual and user-friendly experience but relies on static templates rather than dynamic learning. TechRadar (2022) noted that Lifesum's recommendations do not adjust based on user progress, limiting its flexibility.

To overcome these shortcomings, my project incorporates conversational AI using Dialogflow, a Google platform for building chatbots with natural language understanding (NLU). Dialogflow allows for contextual conversation flow and integration with APIs through Python-based fulfillment.

The primary external data source is Spoonacular, which provides structured access to recipes, nutrition data, and meal plans. It allows filtering by diet type (vegan, keto, low-carb), calories, or allergens. The chatbot uses Spoonacular to retrieve dynamic meal suggestions that match user preferences.

However, in my updated design, I have also integrated fallback APIs such as TheMealDB (for recipes) and Open Food Facts (for nutrition data). These free, open-source APIs ensure functionality even if Spoonacular becomes inaccessible or exceeds budget limits. This multi-source approach builds system reliability and demonstrates strong project risk management.

Academic studies support the importance of personalization. Zhou, Li, and Wang (2022) found that users are more consistent with nutrition goals when AI tools provide adaptive feedback and emotional support. Similarly, Kim and Singh (2021) argued that AI nutrition tools must consider emotional and contextual behavior to be truly effective.

Thus, my project combines lessons from these studies and builds on existing tools with a conversational, adaptive, and API-driven approach to health and nutrition.

III. Analysis and Synthesis

When analyzing why many nutrition apps fail, three themes emerge: lack of personalization, limited feedback, and poor usability. My research and testing confirm these weaknesses and show where an AI chatbot can fill the gaps.

1. Lack of Personalization:

Most apps provide one-size-fits-all advice. Huang, Zhang, and Liu (2023) demonstrated that personalized recommendations significantly improve user engagement and goal success. MyFitnessPal and Lifesum both struggle here; they rely on user logging instead of adaptive learning. The Smart Meal Planner corrects this by analyzing diet filters and returning real-time meal plans from Spoonacular or backup APIs.

2. Limited Feedback:

Traditional apps report data after eating, offering little guidance at the moment. The Smart Meal Planner aims to provide proactive tips and encouragement. For example, if a user skips breakfast, the chatbot can respond: “No worries here’s a high-protein lunch to keep your energy steady.”

3. Accessibility and Engagement:

A friendly conversational tone helps users trust and enjoy the tool. Using Dialogflow’s NLU, I designed intents such as Meal Plan Request and Recipe Search, allowing users to interact naturally (“Plan my meals for the week” or “Find low-carb dinners”).

Privacy and simplicity are also essential. Following World Health Organization (2022) guidance, my chatbot limits data storage and maintains transparency about how information is used.

Finally, cost and scalability were key design factors. Dialogflow and Spoonacular both provide free tiers suitable for prototypes, and Python Flask integration allows cost-effective API fulfillment without deploying full servers.

In short, this analysis guided me to a solution that is not only technically sound but human-centered, affordable, and sustainable.

IV. Proposed Solution

The Automated Smart Meal Planner is a chatbot-based system that helps users plan healthy meals and make informed food choices. It integrates AI-driven conversation with live recipe and nutrition data.

Core Components:

- Dialogflow: Provides the chatbot interface and intent recognition.
- Spoonacular API: Supplies meal plans, nutritional data, and recipe suggestions.
- Python (Flask): Manages webhook fulfillment, processes data, and formats chatbot responses.
- Fallback APIs: TheMealDB and Open Food Facts serve as backups for free and reliable access.

Key Features:

- Personalized daily or weekly meal plans.
- Automatic grocery list generation.
- Real-time recipe filtering by diet type, calories, or allergies.
- Supportive messages and nutrition feedback.
- Privacy protection through minimal data collection.

Budget and Resources:

All main tools (Dialogflow, Spoonacular's free tier, TheMealDB, and Python/Colab) are available at no cost for educational use, making this project feasible without financial barriers.

Staffing:

This is a solo project. I am responsible for all development, testing, and documentation.

The system's design prioritizes adaptability. A data-layer abstraction allows API sources to be swapped easily without rewriting the chatbot logic. This ensures project continuity regardless of external changes.

Implementation Progress (as of October 2025)

Current Achievements:

- Dialogflow intents created: Meal Plan Request and Recipe Search.
- Flask webhook connected successfully to Spoonacular API using Pyngrok.
- One-day meal plan tested and verified with live data.
- Spoonacular responses formatted and displayed through the chatbot interface.
- Backup APIs (TheMealDB and Open Food Facts) tested and verified for fallback integration.

Challenges Overcome:

- Initial confusion about Dialogflow training phrases and webhook setup.
- Managing large API responses and formatting errors during testing.
- Handling Spoonacular's free-tier quota limits by preparing backup API strategies.

Next Steps:

- Expand to multi-day meal planning (3-day and 7-day options).
- Add user-friendly error messages and improve response formatting.
- Implement calorie-tracking summaries and dietary preference settings.
- Begin peer testing for chatbot usability and clarity.

After being sick and falling behind earlier in the semester, I regained momentum by breaking each task into smaller, achievable goals—first connecting Spoonacular, then adding

Dialogflow intents, and finally testing responses. That step-by-step process helped rebuild my confidence and keep the project moving forward despite earlier setbacks.

V. Current Status and Work Plan

This section outlines the remaining milestones for completing the Automated Smart Meal Planner within the next eight weeks. Each task reflects the skills and tools learned throughout my Information Technology program, emphasizing AI integration, API management, and user-focused design.

Weekly Task Plan:

Week	Task	Tools/Skills	Status
1	Finalize Proposal and API Tests	Python, Spoonacular API	Complete
2	Add Calorie and Diet Filters	Dialogflow, Spoonacular	In Progress
3	Connect Dialogflow Flask Webhook	Flask, Pyngrok	In Progress
4	Integrate Backup APIs (TheMealDB / Open Food Facts)	Python, REST	Planned
5	Expand Meal-Plan Logic and Improve Chat Responses	Dialogflow, UX Design	Planned
6	Dialogflow, UX Design	Google Forms	Planned
7	Security, Privacy, and Final Debugging	WHO Guidelines, Python	Planned
8	Final Demo and Prepare/Presentation	Google Slides, Documentation	Planned

This 8-week timeline provides a focused, achievable path toward completing the Smart Meal Planner before finals week. Each task builds logically from the last testing functionality, improving conversation quality, and validating results through feedback and refinement. The final weeks are reserved for presentation prep and quality assurance to ensure a professional, working prototype by the semester's end.

VI. References

Fitbit. (2024). *How Fitbit tracks food and macros*.

<https://www.fitbit.com/global/us/technology/food-logging>

Google Cloud. (2024). *Dialogflow documentation*. <https://cloud.google.com/dialogflow/docs>

Healthline. (2023). *MyFitnessPal review: What to know before using it*.

<https://www.healthline.com/health/myfitnesspal-review>

Huang, Y., Zhang, J., & Liu, S. (2023). Personalized meal planning using AI and nutrition data mining. *Health Informatics Journal*, 29(1), 1–15.

Kim, J., & Singh, A. (2021). Making meal tracking smarter: Emotional and contextual awareness in AI nutrition tools. *ACM Ubiquitous Technologies*, 5(3), 1–23.

Kumar, R., & Patel, A. (2023). Intelligent diet recommendation systems using machine learning: A comparative review. *Computers in Biology and Medicine*, 158, 106808.

Spoonacular. (2024). *Spoonacular Recipe and Food API*. <https://spoonacular.com/food-api>

TechRadar. (2022). *Lifesum app review*. <https://www.techradar.com/reviews/lifesum>

World Health Organization. (2022). *Healthy diet*.

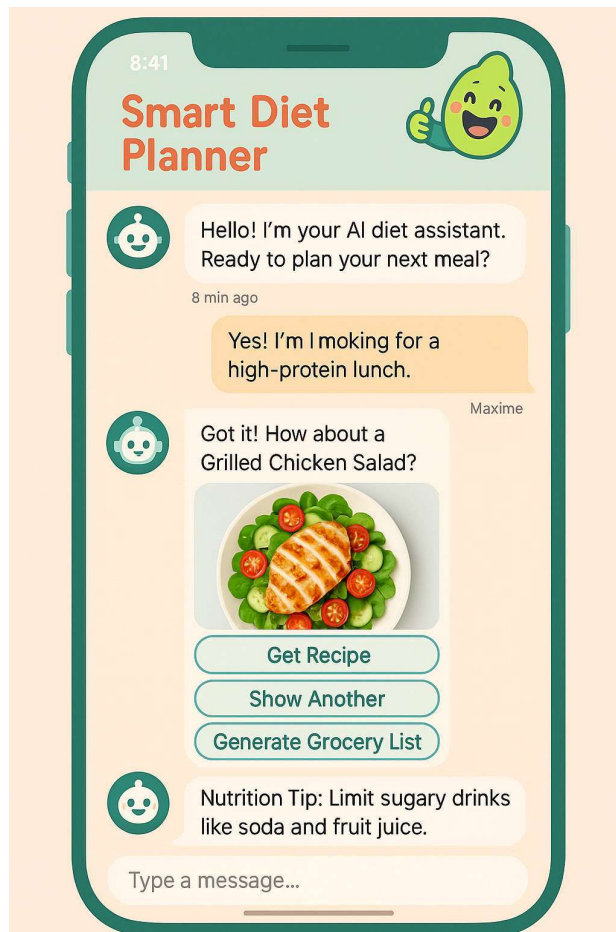
<https://www.who.int/news-room/fact-sheets/detail/healthy-diet>

Zhou, M., Li, Y., & Wang, X. (2022). AI-driven diet coaching improves adherence to nutrition plans: A randomized study. *Journal of Biomedical Informatics*, 128, 104010.

Appendix A

Figure 1

Early Mock-Up of the Smart Diet Planner Chatbot Interface (Concept Design)



The early concept interface displayed a friendly chat window where the user asked, “Can you suggest a healthy lunch?” and the chatbot replied with an example meal plan and grocery list.

The design emphasized approachability, rounded chat bubbles, a simple blue-and-white color scheme, and concise text focused on clarity. The concept illustrates how the final chatbot will guide users through natural conversation while keeping a clean, visually balanced layout.

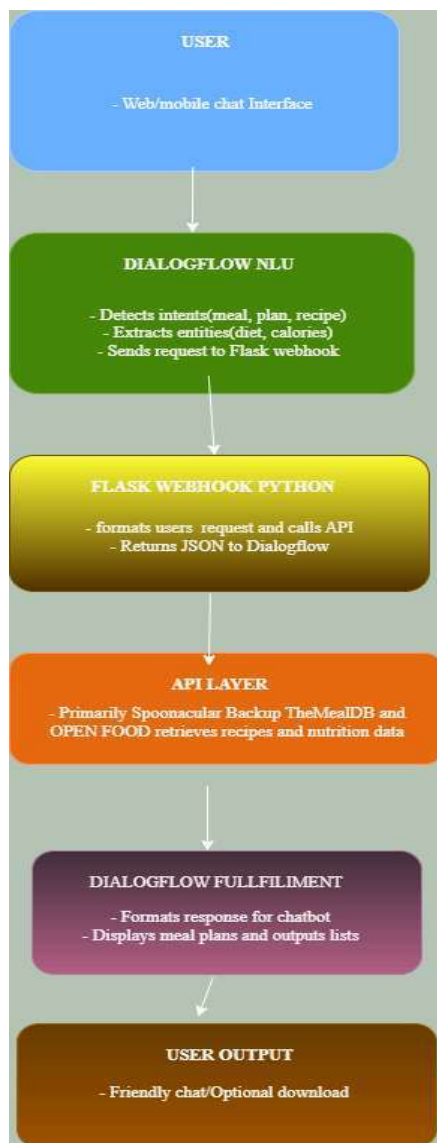
Final chatbot interface design screenshot will be added in the completed project submission.

Appendix B

System Architecture Diagram

Figure 2

Automated Smart Meal Planner System Architecture



Appendix C

Project Timeline (8-Week Plan)

Week	Task	Oct	Nov	Dec
1	Finalize Proposal and API Tests	■ ■		
2	Add Calorie and Diet Filters	■ ■		
3	Connect Dialogflow Flask Webhook	■ ■		
4	Integrate Backup APIs (TheMealDB / Open Food Facts)		■ ■	
5	Expand Meal-Plan Logic and Improve Chat Responses		■ ■	
6	Conduct User Testing and Collect Feedback		■ ■	
7	Security, Privacy, and Final Debugging		■ ■	
8	Final Demo and Presentation Preparation			■ ■

Figure 3

8-Week Development Schedule

Conclusion and Reflection

The Automated Smart Meal Planner represents the intersection of technology, health awareness, and human-centered design. Through this project, I identified the ongoing challenge of making meal planning both intelligent and approachable. My proposal combines conversational AI, reliable nutrition APIs, and a flexible system architecture to create a personalized user experience that supports long-term healthy habits.

By integrating Dialogflow, Spoonacular, and Python Flask, the system demonstrates a cost-effective and scalable approach for interactive meal planning. The design encourages accessibility by using natural language and offers users consistent feedback that adapts to their needs. While the project is still in development, the progress achieved so far especially in building and testing API connections lays a strong foundation for the full implementation phase and the final presentation.

This project has also been a personal journey. Earlier this semester, I faced illness and setbacks that temporarily slowed my progress. Returning to this work required focus, discipline, and self-compassion. Breaking tasks into smaller steps helped me rebuild momentum, and each successful milestone reminded me of my resilience. The experience taught me the importance of patience in development and the value of perseverance in both academics and life. I am proud of

the recovery and growth that this project represents, and I am motivated to see it through to completion.

In conclusion, the Automated Smart Meal Planner is not only a demonstration of technical skill but also a reflection of personal strength and professional readiness. It combines the lessons of my Information Technology program with real-world problem-solving, illustrating how innovation can inspire healthier and more balanced lifestyles.